Executive Summary: Validation of a Mortgage PD Model

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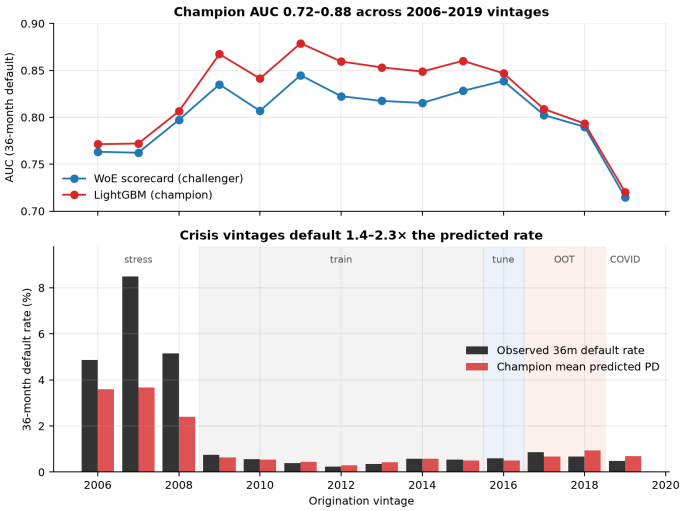
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What was validated. A 36-month probability-of-default model for conforming fixed-rate mortgages at origination: a LightGBM champion with monotone constraints, benchmarked against a weight-of-evidence scorecard and a FICO-only baseline, on Freddie Mac's public loan-level sample (fourteen vintages, 2006–2019, 700,000 loans). Validation follows SR 26-2 (April 2026): conceptual soundness, outcomes analysis and ongoing monitoring, against thresholds set in `configs/default.yaml`.

Intended use. Ranking new originations for screening and review; point-in-time PD within the post-2009 regime. Not for capital, pricing or stress without a macro overlay.

Overall rating: Fit for intended use, with conditions.

	Champion	Scorecard	FICO-only
AUC, OOT 2017–18 (97,449 loans / 755 defaults)	**0.795** (0.780–0.809)	0.790	0.739
Observed/predicted PD, OOT	**0.96**	0.88	1.13
Observed/predicted PD, 2006–08 crisis	**1.9×**	2.3×	5.0×
AUC, crisis vintages	0.783	0.775	0.738



Top findings (1 High, 10 Medium, 2 Low):

- **F10 (High, remediated).** Monotone constraints not enforced with LightGBM's 'advanced' method. Champion uses 'intermediate'.
- **F01 (Medium).** Champion train-to-OOT AUC decay exceeds tolerance. Part in-sample optimism (train-tune gap), part regime drift (tune-OOT). Tighten regularisation; monitor AUC quarterly; re-estimate on rolling window.
- **F02 (Medium).** Weak rank-ordering in segment fico_band=<660. Investigate segment-specific drivers; consider segment overlay.

Conditions of use. Macro overlay or recalibration before any use of PD levels outside 2009–2018; quarterly monitoring of score PSI, AUC, calibration-in-the-large and the share of delinquencies under hardship flags; replace the origination-rate input with a rate spread and complete a fair-lending review with protected-class data before policy use.